Periodic patterns

Arrange & explain

1 Modern & historical tables

Modern order: **atomic number.** Groups link outer electrons and reactions; periods link occupied shells.

Mendeleev used mass/properties, left gaps and predicted missing elements. Discoveries supported predictions; isotopes explain mass-order anomalies.

2 Group 0

Full outer shell: 8 electrons except **helium has 2.** Very unreactive, monatomic gases.

Boiling point increases with atomic mass down the group. Use the trend to predict values.

Compare the groups

3 Group 1

One outer electron → **+1 ions.** Li, Na, K: water → hydroxide + H_2; oxygen → oxides; chlorine → chlorides.

$2Na + 2H_2O \rightarrow 2NaOH + H_2$. Alkaline solution; reactions get more vigorous Li → Na → K.

Down group: more distance/shielding → outer electron lost more easily → reactivity rises.

4 Group 7

Seven outer electrons; gain one → **−1 ions.** Diatomic Cl_2, Br_2, I_2. Metal halides ionic; non-metal compounds covalent.

Cl_2 green gas; Br_2 red-brown liquid; I_2 grey-black solid. Melting/boiling points rise down group.

More distance/shielding → weaker attraction for incoming electron → reactivity falls.

Apply predictions

5 Halogen displacement

More reactive halogen displaces less reactive halogen from its aqueous salt.

$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$. Ionic: $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$.

Iodine cannot displace chlorine from chloride. Use group position to predict.

6 Transition metals

Cr, Mn, Fe, Co, Ni, Cu: generally stronger/harder/denser, higher melting and less reactive than Group 1.

Many have variable ion charges, coloured compounds and catalytic uses.

Fe^{2+}/Fe^{3+}; blue copper sulfate; iron/nickel catalysts. Trends have exceptions.